Q1	/35
Q2	/30
Q3	/30
TOTAL	/100

CSE4088 Introduction to Machine Learning
Fall 2023-2024
FINAL EXAM

Name:
Student ID:

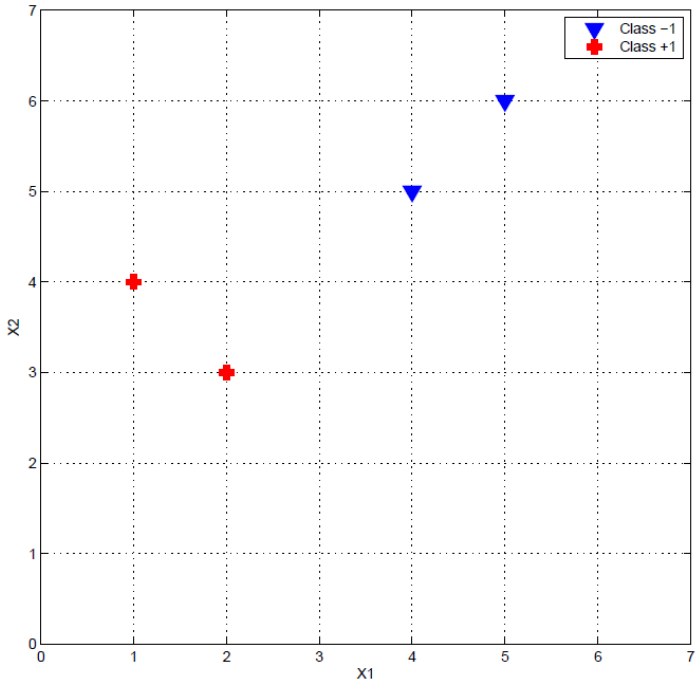
- **NOW please write your name at the top of this page.**
- **There are 3 questions in this exam. Check the pages to make sure that you have all questions.**
- **Turn in ALL the pages at the end of the exam.**
- **You have 90 minutes to complete this exam.**
- **Closed books and notes. You may use a simple calculator.**
- **SHOW ALL YOUR WORK CLEARLY AND STEP BY STEP TO RECEIVE CREDIT. Good luck!**

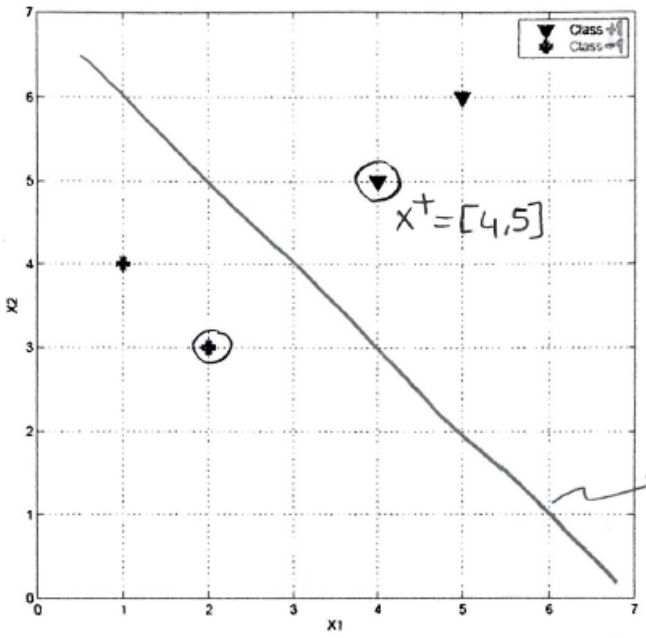
QUESTION 1 – Support Vector Machines

Support vector machines learn a decision boundary with the largest margin from both classes. You are training SVM on a tiny dataset with 4 points shown in the Figure below. This dataset consists of two examples with class label +1 (denoted with plus), and two examples with class label -1 (denoted with triangles).

(i) Find the weight vector w and the bias b. What is the equation corresponding to the decision boundary?

(ii) Circle the support vectors and draw the decision boundary.





Class +1

Class -1

Decision boundary

$\underline{w} \cdot \underline{x} + b = 0$

$w_1 x_1 + w_2 x_2 + b = 0$

$x_1 + x_2 - 7 = 0$

$\underline{w} \cdot \underline{x}^T + b = 1 \Rightarrow \underline{w} \cdot [4,5] + b = 1$

$\underline{w} \cdot \underline{x}^T + b = -1 \Rightarrow \underline{w} \cdot [2,3] + b = -1$

$\frac{1}{2} x_1 + \frac{1}{2} x_2 - \frac{7}{2} = 0$ Decision Boundary

$\underline{w} = \left[\frac{1}{2}, \frac{1}{2}\right]$

$b = \frac{-7}{2}$

QUESTION 2

Suppose a bank classifies customers as either good or bad credit risks. On the basis of extensive historical data, the bank has observed that 1% of good credit risks and 10% of bad credit risks overdraw their account in any given month. A new customer opens a checking account at this bank. On the basis of a check with a credit bureau, the bank believes that there is a 70% chance the customer will turn out to be a good credit risk.

(a) (12 pts) Suppose that this customer's account is overdrawn in the first month. How does this alter the bank's opinion of this customer's creditworthiness?

(b) (13 pts) Given (a), what would be the bank's opinion of the customer's creditworthiness at the end of the second month if there was not an overdraft in the second month?

(**Hint**: Let G and O represent the following events:
G: customer is considered to be a good credit risk.
O: customer overdraws the checking account.
Then, the question provides the information that $P(O|G) = 0.01$ and $P(O|{\sim}G) = 0.1$ and $P(G) = 0.7$)

(a) (5 pts) Suppose that this customer's account is overdrawn in the first month. How does this alter the bank's opinion of this customer's creditworthiness?

Answer:

Let G and O represent the following events:

G: customer is considered to be a good credit risk.

O: customer overdraws checking account.

From the bank's historical data, we have $P(O|G) = 0.01$ and $P(O|\tilde{G}) = 0.1$, and we also know that the bank's initial opinion about the customer's creditworthiness is given by $P(G) = 0.7$.

Given the information that the customer has an overdraft in the first month, the bank's revised opinion about the customer's creditworthiness is given by the conditional probability $P(G|O)$. Using Bayes' theorem and the law of total probability,

$$\begin{aligned} P(G|O) &= \frac{P(O|G)P(G)}{P(O)} \\ &= \frac{P(O|G)P(G)}{P(O|G)P(G) + P(O|\tilde{G})P(\tilde{G})} \\ &= \frac{0.01 \times 0.7}{0.01 \times 0.7 + 0.1 \times 0.3} \\ &\approx 0.189 \end{aligned}$$

(b) (5 pts) Given (a), what would be the bank's opinion of the customer's creditworthiness at the end of the second month if there was not an overdraft in the second month?

Answer:

According to (a), at the beginning of the second month, the prior probability of the customer being a good risk is 0.189. Since the customer does not have an overdraft

in the second month, the bank's opinion about the customer's creditworthiness at the end of the month is given by

$$\begin{aligned} P(G|\tilde{O}) &= \frac{P(\tilde{O}|G)P(G)}{P(\tilde{O})} \\ &= \frac{P(\tilde{O}|G)P(G)}{P(\tilde{O}|G)P(G) + P(\tilde{O}|\tilde{G})P(\tilde{G})} \\ &= \frac{0.99 \times 0.189}{0.99 \times 0.189 + 0.9 \times 0.811} \\ &\approx 0.204 \end{aligned}$$

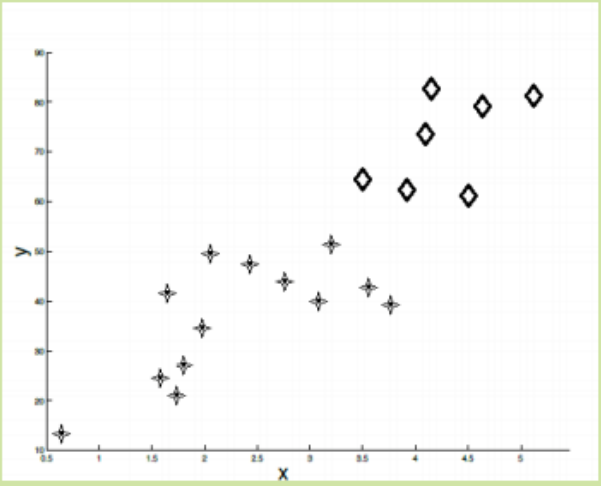
QUESTION 3 – Principal Component Analysis (PCA)

(a) In each of the following plots, a training set of data points X belonging to two classes on $\mathbb{R}^2$ is given, where the original features are the coordinates (x, y). For each plot, answer the following questions:

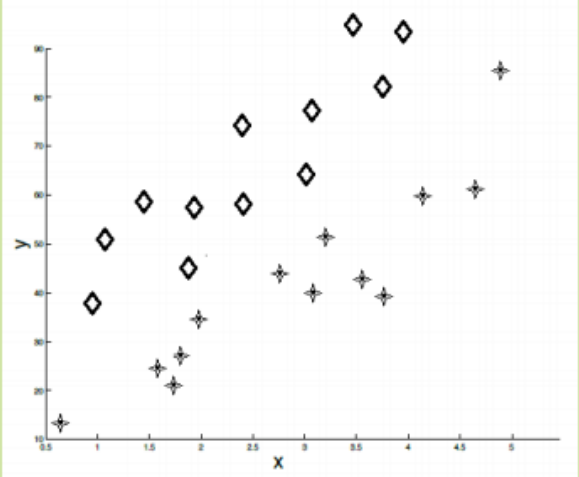
i. Draw all the principal components.

ii. Can we correctly classify this dataset by using a threshold function after projecting onto one of the principal components? If so, which principal component should we project onto? If not, explain in 1-2 sentences why it is not possible.

Dataset 1:



Dataset 2:



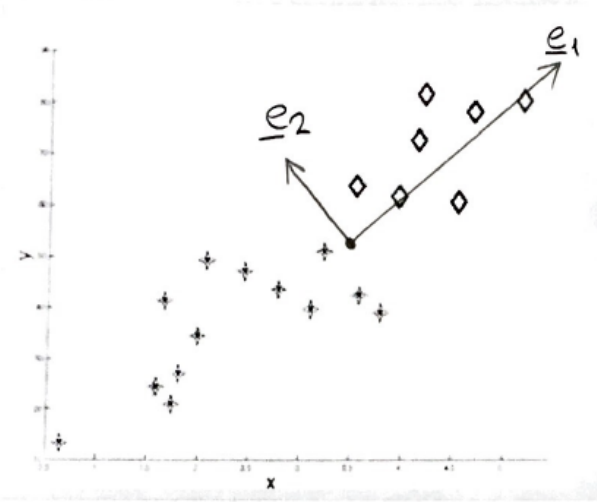
(a) Indicate whether the below statements are **true** or **false**. Explain briefly.

i. (…….) The goal of PCA is to interpret the underlying structure of the data in terms of the principal components that are best at predicting the output variable.

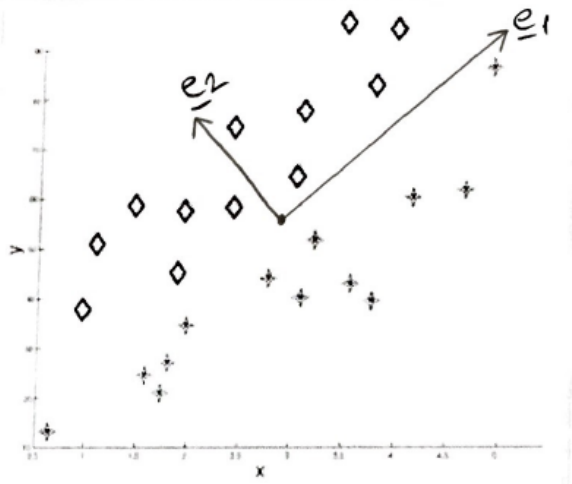
ii. (…….) The output of PCA is a new representtion of the data that is always of lower dimensionality than the original feature representation.

iii. (…….) Subsequent principal components are always orthogonal to each other.

Dataset 1:



Dataset 2:



ii) Dataset 1: Yes. Project the data onto e_1

Dataset 2: Yes. Project the data onto e_2.

(b) Indicate whether the below statements are **true** or **false**. Explain briefly.

i. (..F.....) The goal of PCA is to interpret the underlying structure of the data in terms of the principal components that are best at predicting the output variable.

The goal of PCA is to represent the data in the best possible way. It's goal is not to classify.

ii. (..F.....) The output of PCA is a new representtion of the data that is always of lower dimensionality than the original feature representation.

In PCA we can keep the dimensionaly of the original feature representation the same. The data is decorrelated.

iii. (....T.....) Subsequent principal components are always orthogonal to each other. Since e_i are the eigenvectors of the scatter matrix, which is real and symmetric, they are orthogonal.